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Keyswitch structure

Abstract

A keyswitch structure includes a casing, a first support, a second support, and a pressing stem. The casing forms an accommodating space and an opening communicating with the accommodating space. The first and second supports are disposed in the accommodating space and are directly and rotatably connected with the casing; the supports are also pivotally connected with each other. The pressing stem extends into the accommodating space to be rotatably connected with the first and second supports and protrudes from the casing through the opening. The pressing stem is movable parallel to a vertical direction relative to the casing through the first and second supports. A motion of the pressing stem in the vertical direction has a top dead center and a bottom dead center. When the pressing stem is at the top dead center and the bottom dead center, it does not touch the casing in the vertical direction.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Application No. 63/410,638, filed on Sep. 28, 2022. The content of the application is incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to a mechanical key.

2. Description of the Prior Art

(2) A mechanical key usually includes multiple movable components. When a user presses the key, these movable components will move relative to each other and may collide with each other, producing sound. For example, in general, the upper and lower dead centers of the up and down reciprocating motion of the pressing stem of the key are realized by the casing of the key blocking the up and down motion of the pressing stem; that is, the pressing stem will hit the casing. The

pressing stem hitting the casing produces sound. The harder the user presses on the pressing stem, the louder the pressing stem hits the casing. This phenomenon makes mechanical keys difficult to be used as keys for silent keyboards.

SUMMARY OF THE INVENTION

(3) An objective of the invention is to provide a keyswitch structure, which uses a linkage mechanism to control the locations of the upper and bottom dead centers of the up and down reciprocating motion, which can effectively avoid the hit between components, thereby reducing the sound produced when the keyswitch structure is operated.

(4) A keyswitch structure of an embodiment according to the invention includes a casing, a first support, a second support, and a pressing stem. The casing forms an accommodating space and has an opening communicating with the accommodating space. The first support is disposed in the accommodating space and is directly and rotatably connected with the casing. The second support is disposed in the accommodating space and is directly and rotatably connected with the casing. The first support and the second support are pivotally connected with each other. The pressing stem extends into the accommodating space to be rotatably connected with the first support and the second support and protrudes from the casing through the opening. The pressing stem is movable parallel to a vertical direction relative to the casing through the first support and the second support. A motion of the pressing stem in the vertical direction has a top dead center and a bottom dead center. Therein, when the pressing stem is at the top dead center and the bottom dead center, the pressing stem does not touch the casing in the vertical direction. Thereby, since the pressing stem will not hit the casing, sound produced by hitting the casing can be avoided, which is conducive to silent designs.

(5) These and other objectives of the present invention will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the preferred embodiment that is illustrated in the various figures and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic diagram illustrating a keyswitch structure according to a first embodiment.

(2) FIG. 2 is an exploded view of the keyswitch structure in FIG. 1.

(3) FIG. 3 is a sectional view of the keyswitch structure in FIG. 1 along the line X1-X1.

(4) FIG. 4 is a sectional view of the keyswitch structure in FIG. 3 when the pressing stem of the keyswitch structure is at an intermediate position.

(5) FIG. 5 is a sectional view of the keyswitch structure in FIG. 3 when the pressing stem is at a bottom dead center.

(6) FIG. 6 is a graph of pressing force versus displacement of the pressing stem for the keyswitch structure in FIG. 1.

(7) FIG. 7 is a sectional view of a keyswitch structure according to a second embodiment.

(8) FIG. 8 is a schematic diagram illustrating a keyswitch structure according to a third embodiment.

(9) FIG. 9 is an exploded view of the keyswitch structure in FIG. 8.

(10) FIG. 10 is a sectional view of the keyswitch structure in FIG. 8 along the line X3-X3.

(11) FIG. 11 is a sectional view of the keyswitch structure in FIG. 10 when the pressing stem is at a bottom dead center.

(12) FIG. 12 is a graph of pressing force versus displacement of the pressing stem for the keyswitch structure in FIG. 8.

(13) FIG. 13 is a sectional view of a keyswitch structure according to a fourth embodiment.

- (14) FIG. 14 is a sectional view of a keyswitch structure according to a fifth embodiment.
- (15) FIG. 15 is a schematic diagram illustrating a keyswitch structure according to a sixth embodiment.
- (16) FIG. 16 is an exploded view of the keyswitch structure in FIG. 15.
- (17) FIG. 17 is a sectional view of the keyswitch structure in FIG. 15 along the line X6-X6.
- (18) FIG. 18 is a sectional view of the keyswitch structure in FIG. 17 when the pressing stem is at a bottom dead center.
- (19) FIG. 19 is an exploded view of a keyswitch structure according to a seventh embodiment.
- (20) FIG. 20 is a top view of first and second supports inside the keyswitch structure in FIG. 19 and a lower cover; therein, the first and second supports are in a horizontal posture.
- (21) FIG. 21 is a sectional view of a keyswitch structure according to an eighth embodiment.
- (22) FIG. 22 is a schematic diagram illustrating interference between movement tracks of a first support and a link of the keyswitch structure in FIG. 21.
- (23) FIG. 23 is a sectional view of a keyswitch structure according to a ninth embodiment.
- (24) FIG. 24 is a schematic diagram illustrating interference between movement tracks of a first support and a link of the keyswitch structure in FIG. 23.

DETAILED DESCRIPTION

(25) Please refer to FIG. 1 to FIG. 3. A keyswitch structure 1 according to a first embodiment includes a casing 10, a first support 11, a second support 12, a pressing stem 13, a lateral force generating mechanism 14, and a spring 15. The casing 10 includes an upper cover 102 and a lower cover 104. The upper cover 102 and the lower cover 104 are engaged to form an accommodating space 10a. The upper cover 102 has an opening 10b communicating with the accommodating space 10a. The first support 11 and the second support 12 are disposed in the accommodating space 10a and are pivotally connected with each other relative to a rotation axis A1 (in FIG. 2, indicated by a chain line; in FIG. 3, indicated by a cross mark). The pressing stem 13 extends into the accommodating space 10a to be rotatably connected with the first support 11 and the second support 12, and protrudes from the casing 10 through the opening 10b. Both the first support 11 and the second support 12 are directly and rotatably connected to the casing 10 and the pressing stem 13. Therein, the first support 11 has a pressing stem connecting portion 112 and is pivotally connected with the pressing stem 13 through the pressing stem connecting portion 112. The first support 11 also has a casing connecting portion 114 and is slidably and rotatably connected with the casing 10 (or a sliding slot formed on the lower cover 104) through the casing connecting portion 114. The second support 12 has a pressing stem connecting portion 122 and is slidably and rotatably connected with the pressing stem 13 (or a sliding slot formed on the upper cover 102) through the pressing stem connecting portion 122. The second support 12 also has a casing connecting portion 124 and is pivotally connected with the casing 10 (or the lower cover 104 thereof) through the casing connecting portion 124. The pressing stem 13 is movable parallel to a vertical direction Dv1 (indicated by a double-headed arrow in the figures) relative to the casing 10 through the first support 11 and the second support 12 (i.e., the pressing stem 13 can move up and down).

(26) Furthermore, the lateral force generating mechanism 14 is disposed in the accommodating space 10a and can apply a lateral force F1 to the first support 11. The lateral force F1 is not parallel to the vertical direction Dv1. In the first embodiment, the lateral force F1 is substantially parallel to a horizontal direction Dh1 (indicated by a double-headed arrow in the figures), i.e., perpendicular to the vertical direction Dv1. The lateral force F1 pushes the first support 11 (or the casing connecting portion 114 or a portion of the first support 11 close to the casing connecting portion 114) to make the pressing stem 13 tend to move upward or downward (as described below). The spring 15 is pre-compressed and disposed in the accommodating space 10a. The spring 15 abuts against and between the casing 10 (or the lower cover 104 thereof) and the pressing stem 13 in the vertical direction Dv1. When the pressing stem 13 is pressed (e.g., in practice, a keycap is fixedly

connected to the pressing stem **13**, and a user can press the pressing stem **13** by pressing the keycap), the pressing stem **13** moves downward parallel to the vertical direction **Dv1** through the first support **11** and the second support **12** and squeezes the spring **15**. The resilient force (of which the direction is upward) produced by the squeezed spring **15** (i.e., the elastically deformed spring **15**) can exert an upward force on the pressing stem **13**, so that the spring **15** tends to move upward in the vertical direction **Dv1**.

(27) In the first embodiment, the lateral force generating mechanism **14** may be a bent elastic sheet, which is fixed on the lower cover **104**. In practice, the bent elastic sheet can be replaced with other elastic parts, such as a spiral spring that is disposed horizontally and can also provide the first support **11** a horizontal lateral force. Furthermore, in the first embodiment, the spring **15** is realized by a spiral spring.

(28) In the first embodiment, both the lateral force **F1** (produced by the lateral force generating mechanism **14**) and the resilient force (produced by the spring **15**) affect the up and down movement of the pressing rod **13**. Therein, the lateral force **F1** affects the movement of the pressing stem **13** by generating a moment to the first support **11**; the resilient force is always upward to make the pressing stem **13** tend to move upward. Therefore, the posture of the first support **11** will determine whether the moment makes the pressing stem **13** tend to move upward or downward. Please refer to FIG. 3 to FIG. 5. The pressing stem **13** can move up and down relative to the casing **10** through the first support **11** and the second support **12**. The movement of the pressing stem **13** on the vertical direction **Dv1** can be logically regarded as a reciprocating motion. Therein, when the pressing stem **13** reaches the highest point, the position of pressing stem **13** is defined as a top dead center, as shown by FIG. 3. When the pressing stem **13** reaches the lowest point, the position of pressing stem **13** is defined as a bottom dead center, as shown by FIG. 5. When the moment produced by the lateral force **F1** to the first support **11** is zero, the position of pressing stem **13** is defined as an intermediate position, as shown by FIG. 4; therein, the lateral force **F1** passes through the rotation axis **A1** (indicated by a cross mark in the figures).

(29) Therein, when the pressing stem **13** is at the top dead center (as shown by FIG. 3), the pressing stem connecting portion **112** is higher than the rotation axis **A1** in the vertical direction **Dv1**, and the moment produced by lateral force **F1** on the first support **11** relative to the rotation axis **A1** is defined as a first moment **M1** (indicated by an arc with an arrow in FIG. 3). The first moment **M1** is a clockwise moment (from the view point of the sectional view) and can make the first support **11** tend to move upward; in other words, in the absence of other forces on the pressing stem **13**, the first moment **M1** itself will drive the first support **11** to rotate clockwise (relative to the rotation axis **A1**) to move the pressing stem **13** upward. In the movement of the pressing stem **13** between the top dead center and the intermediate position (i.e., during the movement away from the top center and toward the middle point, or away from the middle point and toward the top dead center), although the first moment **M1** may vary, it is not zero; the closer the pressing stem **13** is to the intermediate position, the closer the first moment **M1** is to zero.

(30) Moreover, when the pressing stem **13** is at the bottom dead center (as shown by FIG. 3), the pressing stem connecting portion **112** is lower than the rotation axis **A1** in the vertical direction **Dv1**, and the moment produced by lateral force **F1** on the first support **11** relative to the rotation axis **A1** is defined as a second moment **M2** (indicated by an arc with an arrow in FIG. 5). The second moment **M2** is a counterclockwise moment and can make the first support **11** tend to move downward; in other words, in the absence of other forces on the pressing stem **13**, the second moment **M2** itself will drive the first support **11** to rotate counterclockwise (relative to the rotation axis **A1**) to move the pressing stem **13** downward. Similarly, in the movement of the pressing stem **13** between the bottom dead center and the intermediate position (i.e., during the movement away from the bottom dead center and toward the middle point, or away from the middle point and toward the bottom dead center), although the second moment **M2** may vary, it is not zero; the closer the pressing stem **13** is to the intermediate position, the closer the second moment **M2** is to

zero.

(31) In the actual operation, the movement of the pressing stem **13** is determined by the resultant force of the external forces it receives. In the first embodiment, the external forces received by the pressing stem **13** mainly include the resilient force of the spring **15**, the upward supporting forces of the first support **11** and the second support **12**, and the pressing force (e.g., by the user's pressing); therein, the lateral force **F1** indirectly affects the pressing stem **13** through the first support **11**. In the first embodiment, the spring **15** provides vertical resilient force. The lateral force generating mechanism **14** can apply rotational forces in different directions to the first support **11** corresponding to different positions of the pressing stem **13** during the process of pressing the pressing stem **13**, thereby creating a discontinuous pressing feeling (i.e., the tactile feedback of the user pressing). Please refer to FIG. 6, is a graph of pressing force versus displacement of the pressing stem **13** for the keyswitch structure **1**; therein, the pressing force is the pressing force of the user (i.e., which is the source of the user's tactile feeling), and the displacement is the downward displacement of the pressing stem **13**. The first curve (shown in solid line), i.e., the upper curve in the figure, corresponds to the case that the pressing stem **13** is pressed to move from the top dead center to the bottom dead center. The second curve (shown in solid line), i.e., the lower curve in the figure, corresponds to the case that the pressing stem **13** rebounds from the bottom dead center to the top dead center. The displacements corresponding to the peaks of the first and second curves are the intermediate positions. Therefore, when the pressing stem **13** passes through the intermediate positions, the user can sense a significant discontinuous feeling.

(32) Furthermore, please refer to FIG. 3 to FIG. 5. In the keyswitch structure **1**, the second support **12** also includes a triggering portion **126**. A switch **16** (whose setting position is indicated by dashed lines) is correspondingly disposed on the lower cover **104**. In the movement of the pressing stem **13** from the top dead center to the bottom dead center (e.g., when the pressing stem **13** is pressed by the user), the triggering portion **126** triggers the switch **16**. For example, the switch **16** may be an optical switch, e.g., including a light transmitter and a light receiver. The triggering is realized by blocking the light emitted from the light transmitter to the light receiver through the triggering portion **126**. However, it is not limited thereto in practice. For example, the switch **16** may be a mechanical switch (such as a tactile button) or a magnetic switch (such as a Hall switch). Furthermore, in practice, the triggering portion **126** can be changed to be disposed on the first support **11** and the switch **16** is disposed accordingly, which will not be described in addition.

(33) Furthermore, in the keyswitch structure **1**, the pressing stem **13** includes a base portion **132** and a column portion **134** extending upward from the base portion **132**. The base portion **132** is located in the accommodating space **10a** and is connected with the first support **11** and the second support **12**. The column portion **134** protrudes from the casing **10** through the opening **10b**. When the pressing stem **13** is at the top dead center and the bottom dead center, the pressing stem **13** does not touch the casing **10** in the vertical direction **Dv1**. In other words, the pressing stem **13** stops at the top dead center and the bottom dead center through the structural constraints of first support **11** and second support **12**, instead of being blocked at the top dead center and the bottom dead center by structural stops of the casing **10** (e.g., stopping the base portion **132**). Therefore, during the entire process of pressing the pressing stem **13**, the pressing stem **13** (or the base portion **132** thereof) will not hit the casing **10**, so the sound produced by hitting the casing **10** can be avoided. This structural configuration is conducive to the silent design. Furthermore, in the keyswitch structure **1**, the pressing stem **13** is spaced from the casing **10**; therein, the base portion **132** is not in contact with the inner wall of the accommodating space **10a**, and the column portion **134** is not in contact with the periphery of the opening **10b**. When the pressing stem **13** moves vertically between the top dead center and the bottom dead center, the pressing stem **13** is kept out of contact with the casing **10** in the horizontal direction **Dh1**. This structural configuration is also conducive to silent designs.

(34) In the keyswitch structure **1**, the lateral force generating mechanism **14** may be an elastic

sheet; however, it is not limited thereto in practice. Please refer to FIG. 7, which is a sectional view of a keyswitch structure 2 according to a second embodiment; the position of the cutting plane is the same as the line X1-X1 in FIG. 1. The keyswitch structure 2 is substantially similar in structure to the keyswitch structure 1. The keyswitch structure 2 uses the reference symbols of the components of the keyswitch structure 1 in principle. For other descriptions of the keyswitch structure 2, please directly refer to the relevant descriptions and figures of the components with the same name in the keyswitch structure 1, which will not be repeated in addition. Compared with the keyswitch structure 1, a lateral force generating mechanism 24 of the keyswitch structure 2 includes two magnets 242 and 244 (simplified as blocks in the figure), which are oppositely disposed on the casing 10 and the first support 11. A repulsive force is generated by and between the magnets 242 and 244 (e.g., realized by arranging the magnets with the same magnetic poles facing each other) to act on the first support 11 as the lateral force F1. Therefore, the lateral force generating mechanism 24 can also have the same effect on the movement of the pressing stem 13 as the lateral force generating mechanism 14, which will not be repeated in addition.

(35) Similarly, in practice, the magnet 242 can be changed to be disposed on the other side of the magnet 244 in the horizontal direction Dh1 (as shown by dashed lines), so that a magnetic attraction force is generated by and between the two magnets (e.g., realized by arranging the magnets with the opposite magnetic poles facing each other) as the lateral force F1. This structural configuration can also have the above effect on the movement of the pressing stem 13. Moreover, in the instance, one of the magnets 242 and 244 can be replaced with an iron part, which can also generate magnetic attraction. The above instances will not be further described.

(36) Please refer to FIG. 8 to FIG. 10. A keyswitch structure 3 according to a third embodiment includes a casing 30, a first support 31, a second support 32, a pressing stem 33, and a lateral force generating mechanism 34. The casing 30 includes an upper cover 302 and a lower cover 304. The upper cover 302 and the lower cover 304 are engaged to form an accommodating space 30a. The upper cover 302 has an opening 30b communicating with the accommodating space 30a. The first support 31 and the second support 32 are disposed in the accommodating space 30a and are pivotally connected with each other relative to a rotation axis A3 (in FIG. 9, indicated by a chain line; in FIG. 10, indicated by a cross mark). The pressing stem 33 extends into the accommodating space 30a to be rotatably connected with the first support 31 and the second support 32, and protrudes from the casing 30 through the opening 30b. Both the first support 31 and the second support 32 are directly and rotatably connected to the casing 30 and the pressing stem 33. Therein, the first support 31 has a pressing stem connecting portion 312 and is pivotally connected with the pressing stem 33 through the pressing stem connecting portion 312. The first support 31 also has a casing connecting portion 314 and is slidably and rotatably connected with the casing 30 (or a sliding slot formed on the lower cover 304) through the casing connecting portion 314. The second support 32 has a pressing stem connecting portion 322 and is slidably and rotatably connected with the pressing stem 33 (or a sliding slot formed on the upper cover 302) through the pressing stem connecting portion 322. The second support 32 also has a casing connecting portion 324 and is pivotally connected with the casing 30 (or the lower cover 304 thereof) through the casing connecting portion 324. The pressing stem 33 is movable parallel to a vertical direction Dv3 (indicated by a double-headed arrow in the figures) relative to the casing 30 through the first support 31 and the second support 32 (i.e., the pressing stem 33 can move up and down).

(37) Furthermore, the lateral force generating mechanism 34 is disposed in the accommodating space 30a and can apply a lateral force F3 to the first support 31. The lateral force F3 is not parallel to the vertical direction Dv3. In the third embodiment, the lateral force F3 is substantially parallel to a horizontal direction Dh3 (indicated by a double-headed arrow in the figures), i.e., perpendicular to the vertical direction Dv3. The lateral force F3 pushes the first support 31 (or the casing connecting portion 314 or a portion of the first support 31 close to the casing connecting portion 314) to make the pressing stem 33 tend to move upward or downward (as described below).

(38) Please refer to FIG. 10 and FIG. 11. The pressing stem 33 can move up and down relative to the casing 30 through the first support 31 and the second support 32. The movement of the pressing stem 33 on the vertical direction Dv3 can be logically regarded as a reciprocating motion. Therein, when the pressing stem 33 reaches the highest point, the position of pressing stem 33 is defined as a top dead center, as shown by FIG. 10. When the pressing stem 33 reaches the lowest point, the position of pressing stem 33 is defined as a bottom dead center, as shown by FIG. 11. During the movement of pressing stem 33, the pressing stem connecting portion 322 is always higher than the rotation axis A3 in the vertical direction Dv3. The lateral force F3 produces a moment M3 (indicated by an arc with an arrow in FIG. 10 and FIG. 11) relative to the rotation axis A3 for the first support 31. The direction of the moment M3 is always clockwise (from the view point of the sectional view), and the moment M3 can make the first support 31 tend to move upward; in other words, in the absence of other forces on the pressing stem 33, the moment M3 itself will drive the first support 31 to rotate clockwise (relative to the rotation axis A3) to move the pressing stem 33 upward.

(39) In the actual operation, the movement of the pressing stem 33 is determined by the resultant force of the external forces it receives. In the third embodiment, the external forces received by the pressing stem 33 mainly include the upward supporting forces of the first support 31 and the second support 32, and the pressing force (e.g., in practice, a keycap is fixedly connected to the pressing stem 33, and a user can press the pressing stem 33 by pressing the keycap); therein, the lateral force F3 indirectly affects the pressing stem 33 through the first support 31. The moment M3 will change with the position of the pressure stem 33, thereby creating a discontinuous pressing feeling. The moment M3 is the product of the lateral force F3 and a lever arm L3. When the pressing stem 33 moves from the top dead center to the bottom dead center, the lever arm L3 gradually decreases, but the lateral force F3 gradually increases, and the change of their product (i.e., the moment M3) is similar to that of the curve in FIG. 12, which will not be described further. Therein, FIG. 12 is a graph of pressing force versus displacement of the pressing stem 33 for the keyswitch structure 3. In FIG. 12, the pressing force is the pressing force of the user (i.e., which is the source of the user's tactile feeling), and the displacement is the downward displacement of the pressing stem 33. The curve (shown in solid line) in the figure corresponds to the case that the pressing stem 33 is pressed to move from the top dead center to the bottom dead center. The peak of the curve occurs between the top dead center and the bottom dead center (e.g., relatively close to the bottom dead center). Thereby, when pressing the stem 33, the user can sense a significant discontinuous feeling. Furthermore, in the embodiment, the trough of the curve occurs close to the bottom dead center; at this moment, the moment M3 is the smallest and close to zero.

(40) Furthermore, please refer to FIG. 9 and FIG. 10. In the keyswitch structure 3, the first support 31 also includes a triggering portion 316. A switch 36 (whose setting position is indicated by dashed lines) is correspondingly disposed on the lower cover 304. In the movement of the pressing stem 33 from the top dead center to the bottom dead center (e.g., when the pressing stem 33 is pressed by the user), the triggering portion 316 triggers the switch 36. For example, the switch 36 may be an optical switch, e.g., including a light transmitter and a light receiver. The triggering is realized by blocking the light emitted from the light transmitter to the light receiver through the triggering portion 316. However, it is not limited thereto in practice. For example, the switch 36 may be a mechanical switch (such as a tactile button) or a magnetic switch (such as a Hall switch). Furthermore, in practice, the triggering portion 316 can be changed to be disposed on the second support 32 and the switch 36 is disposed accordingly, which will not be described in addition.

(41) Furthermore, similar to the keyswitch structure 1, in the keyswitch structure 3, the pressing stem 33 is spaced from the casing 30, so that during the movement of the pressing stem 33 in the vertical direction Dv3, the pressing stem 33 never touches casing 30. Therefore, during the process of the user pressing the keyswitch structure 3, in principle, there will be no sound due to the impact of the components. This structural configuration is conducive to the silent design.

(42) Furthermore, in the third embodiment, the lateral force generating mechanism **34** may be a bent elastic sheet, which is fixed on the lower cover **304**. In practice, the bent elastic sheet can be replaced with other elastic parts. For example, please refer to FIG. **13**, which is a sectional view of a keyswitch structure **4** according to a fourth embodiment; the position of the cutting plane is the same as the line X3-X3 in FIG. **8**. The keyswitch structure **4** is substantially similar in structure to the keyswitch structure **3**. The keyswitch structure **4** uses the reference symbols of the components of the keyswitch structure **3** in principle. For other descriptions of the keyswitch structure **4**, please directly refer to the relevant descriptions and figures of the components with the same name in the keyswitch structure **3**, which will not be repeated in addition. Compared with the keyswitch structure **3**, a lateral force generating mechanism **44** of the keyswitch structure **4** is realized by a tension spring (shown in thick lines in the figure), which is pre-tensioned and set in the accommodating space **30a**. An end of the tension spring is connected to the casing connecting portion **314** of the first support **31**, and the other end of the tension spring is connected to the casing connecting portion **324** of the second support **32** (or the lower cover **304** of the casing **30**). The resilient force of the tension spring is applied to the first support **31** as the lateral force F3. The lateral force generating mechanism **44** can also have the same effect on the movement of the pressing stem **33** as the lateral force generating mechanism **34**, which will not be repeated in addition.

(43) For another example, please refer to FIG. **14**, which is a sectional view of a keyswitch structure **5** according to a fifth embodiment; the position of the cutting plane is the same as the line X3-X3 in FIG. **8**. The keyswitch structure **5** is substantially similar in structure to the keyswitch structure **3**. The keyswitch structure **5** uses the reference symbols of the components of the keyswitch structure **3** in principle. For other descriptions of the keyswitch structure **5**, please directly refer to the relevant descriptions and figures of the components with the same name in the keyswitch structure **3**, which will not be repeated in addition. Compared with the keyswitch structure **3**, a lateral force generating mechanism **54** of the keyswitch structure **5** includes two magnets **542** and **544** (simplified as blocks in the figure), which are oppositely disposed on the casing **30** and the first support **31**. A repulsive force is generated by and between the magnets **542** and **544** (e.g., realized by arranging the magnets with the same magnetic poles facing each other) to act on the first support **31** as the lateral force F3. Therefore, the lateral force generating mechanism **54** can also have the same effect on the movement of the pressing stem **33** as the lateral force generating mechanism **34**, which will not be repeated in addition.

(44) Similarly, in principle, the magnet **542** can be changed to be disposed on the other side of the magnet **544** in the horizontal direction Dh3 (as shown by dashed lines), so that a magnetic attraction force is generated by and between the two magnets (e.g., realized by arranging the magnets with the opposite magnetic poles facing each other) as the lateral force F3. This structural configuration can also have the above effect on the movement of the pressing stem **33**. Moreover, in the instance, one of the magnets **542** and **544** can be replaced with an iron part, which can also generate magnetic attraction. The above instances will not be further described.

(45) In addition, although the aforementioned keyswitch structures **3**, **4** and **5** do not use the spring **15** in the keyswitch structure **1** as the driving force for the pressing stem **33** to restore, in practice, a spring can also be pre-compressed and disposed in the keyswitch structures **3**, **4** and **5**. The spring abuts against and between the casing **30** (or the lower cover **304** thereof) and the pressing stem **33** in the vertical direction Dv3. This spring has the same function as the spring **15** in keyswitch structure **1**, which will not be repeated in addition.

(46) Please refer to FIG. **15** to FIG. **17**. A keyswitch structure **6** according to a sixth embodiment includes a casing **60**, a first support **61**, a second support **62**, a pressing stem **63**, and a spring **65**. The casing **60** includes an upper cover **602** and a lower cover **604**. The upper cover **602** and the lower cover **604** are engaged to form an accommodating space **60a**. The upper cover **602** has an opening **60b** communicating with the accommodating space **60a**. The first support **61** and the

second support **62** are disposed in the accommodating space **60a** and are pivotally connected with each other relative to a rotation axis **A6** (in FIG. **16**, indicated by a chain line; in FIG. **17**, indicated by a cross mark). The pressing stem **63** extends into the accommodating space **60a** to be rotatably connected with the first support **61** and the second support **62**, and protrudes from the casing **60** through the opening **60b**. Both the first support **61** and the second support **62** are directly and rotatably connected to the casing **60** and the pressing stem **63**. Therein, the first support **61** has a pressing stem connecting portion **612** and a casing connecting portion **614**. The first support **61** is slidably and rotatably connected with the pressing stem **63** through the pressing stem connecting portion **612**, and is pivotally connected with the casing **60** (or the lower cover **604** thereof) through the casing connecting portion **614**. The second support **62** has a pressing stem connecting portion **622** and a casing connecting portion **624**. The second support **62** is slidably and rotatably connected with the pressing stem **63** and the casing **60** (or the lower cover **604** thereof) via the pressing stem connecting portion **622** and the casing connecting portion **624**, respectively. The pressing stem **63** is movable parallel to a vertical direction **Dv6** (indicated by a double-headed arrow in the figures) relative to the casing **60** through the first support **61** and the second support **62** (i.e., the pressing stem **63** can move up and down).

(47) The spring **65** is pre-compressed and disposed in the accommodating space **60a** and abuts against and between the casing **60** (or the lower cover **604** thereof) and the pressing stem **63** in the vertical direction **Dv6**. When the pressing stem **63** is pressed (e.g., in practice, a keycap is fixedly connected to the pressing stem **63**, and a user can press the pressing stem **63** by pressing the keycap), the pressing stem **63** moves downward parallel to the vertical direction **Dv6** through the first support **61** and the second support **62** and squeezes the spring **65**. The resilient force (of which the direction is upward) produced by the squeezed spring **65** (i.e., the elastically deformed spring **65**) can exert an upward force on the pressing stem **63**, so that the spring **65** tends to move upward in the vertical direction **Dv6**.

(48) Please refer to FIG. **17** and FIG. **18**. The pressing stem **63** can move up and down relative to the casing **60** through the first support **61** and the second support **62**. The movement of the pressing stem **63** on the vertical direction **Dv6** can be logically regarded as a reciprocating motion. Therein, when the pressing stem **63** reaches the highest point, the position of pressing stem **63** is defined as a top dead center, as shown by FIG. **17**. When the pressing stem **63** reaches the lowest point, the position of pressing stem **63** is defined as a bottom dead center, as shown by FIG. **18**. In the keyswitch structure **6**, the casing **60** also has a fixed interference portion **6042** on the lower cover **604** in the accommodating space **60a**. The first support **61** has a protruding portion **616**. When the pressing stem **63** is at the top dead center and the bottom dead center, the protruding portion **616** does not touch the fixed interference portion **6042**. In the movement of the pressing stem **63** moving from the top dead center to the bottom dead center, the protruding portion **616** produces elastic structural interference with the fixed interference portion **6042**, so that the fixed interference portion **6042** can generate resistance to the protruding portion **616** to obstruct the clockwise rotation of the first support **61** from rotating clockwise (from the view point of the sectional view). On the other hand, the elastic structural interference between the protruding portion **616** and the fixed interference portion **6042** can create a discontinuous pressing feeling.

(49) Furthermore, in the keyswitch structure **6**, the first support **61** also includes a triggering portion **618**. A switch **66** (whose setting position is indicated by dashed lines) is correspondingly disposed on the lower cover **604**. In the movement of the pressing stem **63** from the top dead center to the bottom dead center (e.g., when the pressing stem **63** is pressed by the user), the triggering portion **618** triggers the switch **66**. For example, the switch **66** may be an optical switch, e.g., including a light transmitter and a light receiver. The triggering is realized by blocking the light emitted from the light transmitter to the light receiver through the triggering portion **618**. However, it is not limited thereto in practice. For example, the switch **66** may be a mechanical switch (such as a tactile button) or a magnetic switch (such as a Hall switch). Furthermore, in practice, the

triggering portion **618** can be changed to be disposed on the second support **62** and the switch **66** is disposed accordingly, which will not be described in addition.

(50) Furthermore, similar to the keyswitch structure **1**, in the keyswitch structure **6**, the pressing stem **63** is spaced from the casing **60**, so that during the movement of the pressing stem **63** in the vertical direction **Dv6**, the pressing stem **63** never touches casing **60**. Therefore, during the process of the user pressing the keyswitch structure **6**, in principle, there will be no sound due to the impact of the components. This structural configuration is conducive to the silent design.

(51) The keyswitch structure **6** uses the elastic structure interference between the first support **61** and the casing **60** to provide a discontinuous pressing feeling; however, it is not limited thereto in practice. For example, please refer to FIG. **19**, which is an exploded view of a keyswitch structure **7** according to a seventh embodiment. The keyswitch structure **7** is substantially similar in structure to the keyswitch structure **6**. The keyswitch structure **7** uses the reference symbols of the components of the keyswitch structure **6** in principle. For other descriptions of the keyswitch structure **7**, please directly refer to the relevant descriptions and figures of the components with the same name in the keyswitch structure **6**, which will not be repeated in addition. Compared with the keyswitch structure **6**, the keyswitch structure **7** uses the elastic deformation of its supports to provide a discontinuous pressing feeling. Please also refer to FIG. **20**, which is a top view of first and second supports **71** and **72** inside the keyswitch structure **7** and the lower cover **604**; therein, the first and second supports **71** and **72** are in a horizontal posture. In the seventh embodiment, the pressing stem **63** of the keyswitch structure **7** also moves up and down relative to the casing **60** through the first support **71** and the second support **72**, logically forming a reciprocating motion. Please refer to FIG. **17** and FIG. **18** for the sectional view of the keyswitch structure **7** when the pressing stem **63** is at the top dead center and the bottom dead center, which will not be shown by another figure. When the pressing stem **63** is at the top dead center and the bottom dead center, the first support **71** and the second support **72** are not deformed. In the process of pressing the pressing stem **63** from the top dead center to the bottom dead center, the second support **72** is elastically deformed. Therein, in principle, when the first support **71** and the second support **72** are in a horizontal posture, the elastic deformation of the second support **72** reaches a maximum, as shown by FIG. **20**. In the seventh embodiment, the elastic deformation of the second support **72** is realized by designing the slot on the pressing stem **63** (i.e., the pressing stem connecting portion **622** that is connected with the second support **72**), such as the length of the slot. When the second support **72** rotates and gradually approaches the horizontal posture, the second support **72** starts to be elastically deformed by bending and gradually increases the degree of bending. This elastic deformation provides a discontinuous pressing feeling. In practice, it is practicable to design the first support **71** to be elastically deformed to provide a discontinuous pressing feeling. It is also practicable to design both the first support **71** and the second support **72** to be elastically deformed to jointly provide a discontinuous pressing feeling.

(52) In practice, one or more of the designs of discontinuous pressing feeling in the above embodiments can be implemented in the same keyswitch structure, which will not be repeated further. In addition, in the above embodiments, the top and bottom dead centers of the pressing stems **13**, **33** and **63** are determined by the action of the mechanisms formed mainly by the connection relationship of the first and second supports **11**, **12**, **31**, **32**, **61**, **62**, **71** and **72** with the pressing stem **13**, **33** and **63** and the casing **10**, **30** and **60**. In practice, constraints on the movement of other components can be added to the above mechanisms to help prevent the pressing stems **13**, **33** and **63** from hitting the casings **10**, **30** and **60**. For example, please refer to FIG. **21**, which is a sectional view of a keyswitch structure **8** according to an eighth embodiment; the position of the cutting plane is the same as the line **X1-X1** in FIG. **1**. The keyswitch structure **8** is substantially similar in structure to the keyswitch structure **1**. The keyswitch structure **8** uses the reference symbols of the components of the keyswitch structure **1** in principle. For other descriptions of the keyswitch structure **8**, please directly refer to the relevant descriptions and figures of the

components with the same name in the keyswitch structure **1**, which will not be repeated in addition. Compared with the keyswitch structure **1**, the keyswitch structure **8** further includes a link **82** (shown by a simple thick line in the figure) which is pivotally connected to the casing **10** (or the lower cover **104** thereof) in the accommodating space **10a**. The link **82** (or an end portion thereof) is pivotally connected with the casing connecting portion **114** of the first support **11**. Please also refer to FIG. **22**, which is a schematic diagram illustrating interference between movement tracks of the link **82** and the casing connecting portion **114**. The movement track **114a** of the casing connecting portion **114** is a horizontal straight line (indicated by a straight dashed line with double arrows in the figure). The movement track **82a** of the end portion of the link **82** connected with the casing connecting portion **114** in principle an arc (indicated by a curved chain line with double arrows in FIG. **22**). The two movement tracks **114a** and **82a** have two intersection points. In principle, the end portion of the link **82** and the casing connecting portion **114** only move between the two intersection points, and then the link **82** limits the movement range of the casing connecting portion **114**, indirectly controlling the locations of the upper and bottom dead centers of the pressing stem **13**. The control of the top and bottom dead centers is not achieved by directly blocking the casing connecting portion **114** with a structure (such as the closed end of the slot on the lower cover **104**), so when the pressing stem **13** reaches the top and bottom dead centers, the casing connecting portion **114** will not hit the lower cover **104** to produce impact sound. Therefore, this structural configuration can further reduce the sound when the keyswitch structure **8** is in operation (relative to the keyswitch structure **1**).

(53) In addition, in the keyswitch structure **8**, in logic, the movement track of the end portion of the link **82** and the movement track of the casing connecting portion **114** do not exactly coincide between the two intersection points, so in practice, the link **82** can be designed to be elastically deformable, so that the link **82** can be bent and deformed so that the end portion thereof and the casing connecting portion **114** can actually move in a straight line (consistent with the movement track of the casing connecting portion **114**). Besides, in the keyswitch structure **8**, the resilient force generated by the elastically deformed link **82** can be designed not to substantially affect the movement of other components (e.g., the lateral force generating mechanism **14**). However, it is not limited thereto in practice. For example, the resilient force of the link **82** is also considered in the design of the discontinuous pressing feeling.

(54) Furthermore, for another example, please refer to FIG. **23**, which is a sectional view of a keyswitch structure **9** according to a ninth embodiment; the position of the cutting plane is the same as the line X1-X1 in FIG. **1**. The keyswitch structure **9** is substantially similar in structure to the keyswitch structure **1**. The keyswitch structure **9** uses the reference symbols of the components of the keyswitch structure **1** in principle. For other descriptions of the keyswitch structure **9**, please directly refer to the relevant descriptions and figures of the components with the same name in the keyswitch structure **1**, which will not be repeated in addition. Compared with the keyswitch structure **1**, the keyswitch structure **9** further includes a link **92** (shown by a simple thick line in the figure), and a first support **91** of the keyswitch structure **9** further includes an extension portion **916** extending from the casing connecting portion **114**. The link **92** is pivotally connected to the casing **10** (or the lower cover **104** thereof) in the accommodating space **10a**. The link **92** (or an end portion thereof) is pivotally connected with the extension portion **916** of the first support **91**. Please also refer to FIG. **24**, which is a schematic diagram illustrating interference between movement tracks of the link **92** and the extension portion **916**. The movement track **916a** of the extension portion **916** (connected with the link **92**) is an arc (indicated by a curved dashed line with double arrows in figure). The end portion of the link **92** connected with the extension portion **916** is also an arc (indicated by a curved chain line with double arrows in the figure). The two movement tracks **916a** and **92a** have two intersection points. In principle, the end portion of the link **92** and the casing connecting portion **916** only move between the two intersection points, and then the link **92** limits the movement range of the casing connecting portion **916**, indirectly controlling the

locations of the upper and bottom dead centers of the pressing stem **13**. Similarly (compared with the keyswitch structure **8**), the control of the top and bottom dead centers is not achieved by directly blocking the casing connecting portion **114** with a structure (such as the closed end of the slot on the lower cover **104**), so when the pressing stem **13** reaches the top and bottom dead centers, the casing connecting portion **114** will not hit the lower cover **104** to produce impact sound. Therefore, this structural configuration can further reduce the sound when the keyswitch structure **9** is in operation (relative to the keyswitch structure **1**).

(55) Furthermore, in the keyswitch structure **9**, in logic, the movement track of the end portion of the link **92** and the movement track of the casing connecting portion **914** do not exactly coincide between the two intersection points, so in practice, the link **92** can be designed to be elastically deformable, so that the link **92** can be bent and deformed so that the end portion thereof and the casing connecting portion **114** can actually move in an arc (consistent with the movement track of the casing connecting portion **114**). Besides, in the keyswitch structure **9**, the resilient force generated by the elastically deformed link **92** can be designed not to substantially affect the movement of other components (e.g., the lateral force generating mechanism **14**). However, it is not limited thereto in practice. For example, the resilient force of the link **92** is also considered in the design of the discontinuous pressing feeling.

(56) Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the invention. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

Claims

1. A keyswitch structure, comprising: a casing, the casing forming an accommodating space and having an opening communicating with the accommodating space; a first support, the first support being disposed in the accommodating space and being directly and rotatably connected with the casing; a second support, the second support being disposed in the accommodating space and being directly and rotatably connected with the casing, the first support and the second support being pivotally connected with each other; and a pressing stem, the pressing stem extending into the accommodating space to be rotatably connected with the first support and the second support and protruding from the casing through the opening, the pressing stem being movable parallel to a vertical direction relative to the casing through the first support and the second support, a motion of the pressing stem in the vertical direction having a top dead center and a bottom dead center, wherein when the pressing stem is at the top dead center and the bottom dead center, the pressing stem does not touch the casing in the vertical direction.

2. The keyswitch structure according to claim 1, wherein when the pressing stem moves vertically between the top dead center and the bottom dead center, the pressing stem is kept out of contact with the casing in a horizontal direction.

3. The keyswitch structure according to claim 1, further comprising a lateral force generating mechanism, disposed in the accommodating space, wherein the lateral force generating mechanism comprises an elastic part, the lateral force generating mechanism applies a lateral force to the first support through the elastic part, the first support and the second support are pivotally connected relative to a rotation axis, the first support has a pressing stem connecting portion and is connected to the pressing stem through the pressing stem connecting portion, wherein when the pressing stem is at the top dead center, the pressing stem connecting portion is higher than the rotation axis in the vertical direction, the lateral force produces a first moment relative to the rotation axis for the first support, and the first moment makes the first support tend to move upward, and wherein when the pressing stem is at the bottom dead center, the pressing stem connecting portion is lower than the rotation axis in the vertical direction, the lateral force produces a second moment relative to the

rotation axis for the first support, and the second moment make the first support tend to move downward.

4. The keyswitch structure according to claim 3, wherein the lateral force is not parallel to the vertical direction.

5. The keyswitch structure according to claim 1, further comprising a spring, disposed in the accommodating space and abutting against and between the casing and the pressing stem in the vertical direction.

6. The keyswitch structure according to claim 1, further comprising a switch, disposed on the casing, wherein the first support or the second support comprises a triggering portion, and in a movement of the pressing stem moving from the top dead center to the bottom dead center, the triggering portion triggers the switch.
